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clc; clear;
% Line Data: [From To R X]
line_data = [
    1 2 0.07 0.15;
    1 3 0.06 0.10;
    1 4 0.08 0.25;
    2 4 0.04 0.10;
    3 4 0.04 0.20
];

n_bus = 4;

% Build Ybus matrix (Using Function)
Ybus = formYbus(line_data, n_bus);
G = real(Ybus);
B = imag(Ybus);

disp('Ybus Matrix (p.u.):');
disp(Ybus);

% Bus Data
V = [1.05; 1.0; 1.0; 1.0]; % initial voltage magnitudes
delta = [0; 0; 0; 0]; % initial phase angles in radians
P = [0; -0.4; -0.5; -0.7]; % injected real powers
Q = [0; -0.15; -0.4; -0.2]; % injected reactive powers

% Slack Bus = 1
% PV Bus = 2
% PQ Buses = 3,4

% Indices
PQ = [3 4];
PV = [2];
npq = length(PQ);
npv = length(PV);

% Step 3: Calculate Jacobian
n = n_bus;
J1 = zeros(n-1, n-1);
J2 = zeros(n-1, npq);
J3 = zeros(npq, n-1);
J4 = zeros(npq, npq);

% J1: dP/dDelta
for i = 2:n
    for j = 2:n
        if i == j
            for k = 1:n
                J1(i-1, j-1) = J1(i-1, j-1) + V(i)*V(k)*(-G(i, k)*sin(delta(i)-
delta(k)) + B(i, k)*cos(delta(i)-delta(k)));
            end
        end
    end
end

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        J1(i-1,j-1) = J1(i-1,j-1) - V(i)^2*B(i,i);
    else
        J1(i-1,j-1) = V(i)*V(j)*(G(i,j)*sin(delta(i)-delta(j)) -
B(i,j)*cos(delta(i)-delta(j)));
    end
end
end

% J2: dP/dV
for i = 2:n
    for j = 1:npq
        m = PQ(j);
        if i == m
            for k = 1:n
                J2(i-1,j) = J2(i-1,j) + V(k)*(G(i,k)*cos(delta(i)-delta(k))
+ B(i,k)*sin(delta(i)-delta(k)));
            end
            J2(i-1,j) = J2(i-1,j) + V(i)*G(i,i); % Fixed formula logic:
usually P_calc/V + Gii*V
        else
            J2(i-1,j) = V(i)*(G(i,m)*cos(delta(i)-delta(m)) +
B(i,m)*sin(delta(i)-delta(m)));
        end
    end
end

% J3: dQ/dDelta
for i = 1:npq
    m = PQ(i);
    for j = 2:n
        if m == j
            for k = 1:n
                J3(i,j-1) = J3(i,j-1) + V(m)*V(k)*(G(m,k)*cos(delta(m)-
delta(k)) + B(m,k)*sin(delta(m)-delta(k)));
            end
            J3(i,j-1) = -J3(i,j-1) - V(m)^2*G(m,m); % Negative sign
adjustment per standard formulas
        else
            J3(i,j-1) = -V(m)*V(j)*(G(m,j)*cos(delta(m)-delta(j)) +
B(m,j)*sin(delta(m)-delta(j)));
        end
    end
end

% J4: dQ/dV
for i = 1:npq
    m = PQ(i);
    for j = 1:npq
        nbus = PQ(j);
        if m == nbus
            for k = 1:n
                J4(i,j) = J4(i,j) + V(k)*(G(m,k)*sin(delta(m)-delta(k)) -
B(m,k)*cos(delta(m)-delta(k)));
            end

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        J4(i,j) = J4(i,j) - V(m)*B(m,m);
    else
        J4(i,j) = V(m)*(G(m,nbus)*sin(delta(m)-delta(nbus)) -
B(m,nbus)*cos(delta(m)-delta(nbus)));
    end
end
end

% Combine Jacobian
J = [J1 J2; J3 J4];
disp('Jacobian Matrix (Initial State):');
disp(J);

Ybus Matrix (p.u.):
    8.1276 -16.4558i  -2.5547 + 5.4745i  -4.4118 + 7.3529i  -1.1611 + 3.6284i
   -2.5547 + 5.4745i    6.0030 -14.0951i    0.0000 + 0.0000i  -3.4483 + 8.6207i
   -4.4118 + 7.3529i    0.0000 + 0.0000i    5.3733 -12.1606i  -0.9615 + 4.8077i
   -1.1611 + 3.6284i  -3.4483 + 8.6207i   -0.9615 + 4.8077i    5.5709 -17.0568i

Jacobian Matrix (Initial State):
    14.3689         0   -8.6207         0   -3.4483
         0    12.5283   -4.8077    5.1527   -0.9615
   -8.6207   -4.8077   17.2383   -0.9615    5.5129
         0   -5.1527    0.9615   11.7930   -4.8077
    3.4483    0.9615   -5.5129   -4.8077   16.8754

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